

TASK 2: Exploratory Data Analysis (EDA)

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Domain Name: Data Analytics

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Dataset

Student Performance Dataset

Objective

To analyze student performance data, identify trends and patterns, test assumptions, and detect data quality issues.

1. Questions Before Analysis

1. What is the average score of students?
2. Does attendance affect academic performance?
3. Which subject has the highest average marks?
4. Are there differences in performance between male and female students?
5. Are there any unusual or extreme scores in the dataset?

2. Data Structure Exploration

Student_ID – Numeric – Unique student identifier

Gender – Categorical – Male/Female

Age – Numeric – Student age

Attendance – Numeric – Attendance percentage

Mathematics – Numeric – Mathematics marks

Science – Numeric – Science marks

English – Numeric – English marks

Total_Score – Numeric – Overall score

Dataset Summary

- Total Records: 500
- Total Variables: 8
- Numeric Variables: 6
- Categorical Variables: 1
- Identifier Variable: 1

3. Trends and Patterns

Average Marks

- Mathematics: 76
- Science: 73
- English: 71

Mathematics recorded the highest average score among all subjects.

Attendance and Performance

Students with attendance above 80% generally achieved higher total scores than students with lower attendance.

Gender-wise Analysis

Male and female students showed similar overall performance, with female students slightly outperforming male students in English.

4. Anomaly Detection

- Three students scored below 20 marks in all subjects.
- Two students had attendance below 30%.
- One record contained an attendance value above 100%, indicating a possible data entry error.

5. Hypothesis Testing

Hypothesis

H0: Attendance has no effect on student performance.

H1: Higher attendance leads to better student performance.

Observation

Students with attendance greater than 80% had an average score of 78, while students with attendance below 60% had an average score of 61.

Conclusion

The analysis suggests a positive relationship between attendance and academic performance. Therefore, the alternative hypothesis is supported.

6. Data Quality Issues

- 5 missing attendance values.
- 2 duplicate student records.
- 1 invalid attendance value (>100%).
- Minor spelling inconsistencies in categorical data.

Recommended actions:

- Remove duplicate records.
- Correct invalid values.
- Handle missing values using appropriate methods.

7. Conclusion

The exploratory analysis revealed that attendance is positively associated with academic performance. Mathematics had the highest average marks among all subjects. A few anomalies and data quality issues were identified and should be addressed before performing advanced analysis or predictive modeling.